# SUPPLEMENTARY MATERIAL

## Adaptive Nonlinear Vector Autoregression: Robust Forecasting for Noisy Chaotic Time Series

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## Supplementary Example 1

We illustrate our adaptive NVAR model using a simple 3-dimensional time series with delay embedding and joint feature learning.

(*i*) Dimension of the state vector: $d = 3$.
(*ii*) Number of delays: $k = 2$.
(*iii*) Neural network output dimension: $d_{\mathcal{NN},i} = dk(dk+1)/2$, to match the number of fixed nonlinear features in NVAR. Alternatively, $d_{\mathcal{NN},i} = c\,dk$, with $c$ being a tunable constant.
(*iv*) Input dimension: $dk + d_{\mathcal{NN},i} = 6 + 21 = 27$.
(*v*) Number of training points: $n = 5$ ($I_{\text{train}} = \{2, 3, 4, 5, 6\}$ are valid training indices for $k = 2$).
(*vi*) For each $i \in I_{\text{train}}$, we use $H_{\text{lin},i} = X_i \oplus X_{i-1} \in \mathbb{R}^6$.

Let $X_i \in \mathbb{R}^d$ for $i = \{1, \ldots, 10\}$ be our toy series dataset, where $d = 3$. The following table lists the components of $X_i$.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $x_{1,i}$ | 0.1 | 0.2 | 0.3 | 0.5 | 0.4 | 0.7 | 0.6 | 0.8 | 0.9 | 0.5 |
| $x_{2,i}$ | 0.3 | 0.1 | 0.6 | 0.2 | 0.5 | 0.1 | 0.4 | 0.6 | 0.5 | 0.2 |
| $x_{3,i}$ | 0.2 | 0.4 | 0.1 | 0.3 | 0.6 | 0.2 | 0.3 | 0.2 | 0.9 | 0.2 |

Supplementary Table 1: Values of $x_{j,i}$ for $j = 1, 2, 3$ and $i = 1, \ldots, 10$.

We now construct the linear feature vectors and their corresponding targets for training:

| $i$ | $H_{\text{lin},i}$ | Target $X_{i+1}$ |
|---|---|---|
| 2 | $X_2 \oplus X_1$ | $X_3$ |
| 3 | $X_3 \oplus X_2$ | $X_4$ |
| 4 | $X_4 \oplus X_3$ | $X_5$ |
| 5 | $X_5 \oplus X_4$ | $X_6$ |
| 6 | $X_6 \oplus X_5$ | $X_7$ |

Supplementary Table 2: Linear combination inputs $H_{\text{lin},i}$ and their target outputs $X_{i+1}$.

The feature vectors are then in the following vector spaces:

1. The linear feature vector $H_{\text{lin},i} \in \mathbb{R}^{dk} = \mathbb{R}^6$.
2. The neural network (nonlinear) feature vector $H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) \in \mathbb{R}^{d_{\mathcal{NN},i}} = \mathbb{R}^{21}$.
3. The total feature vector, defined by concatenating the linear and nonlinear features $H_{\text{total},i} = H_{\text{lin},i} \oplus H_{\mathcal{NN},i} \in \mathbb{R}^{dk + d_{\mathcal{NN},i}} = \mathbb{R}^{27}$.

The neural network $\mathcal{NN}(H_{\text{lin},i}; \theta)$ is defined as a simple MLP with one hidden layer and a tanh activation function. Let the hidden layer have $h = 24$ units. The MLP transformation is given by:

$$H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) = W \cdot \tanh(W_{\text{in}} H_{\text{lin},i}),$$

where $W_{\text{in}} \in \mathbb{R}^{h \times (dk)} = \mathbb{R}^{24 \times 6}$ and $W \in \mathbb{R}^{d_{\mathcal{NN},i} \times h} = \mathbb{R}^{21 \times 24}$.

After concatenating $H_{\text{lin},i}$ and $H_{\mathcal{NN},i}$ into $H_{\text{total},i}$, we perform the readout layer as follows:

$$\hat{Y}_i = \hat{X}_{i+1} - \hat{X}_i = W_{\text{out}} H_{\text{total},i}.$$

Thus, the parameter set $\theta$ is $\theta = \{W_{\text{in}}, W, W_{\text{out}}\}$.

The mean squared error loss

$$L(\theta) = \frac{1}{5} \sum_{i=2}^{6} \|W_{\text{out}} H_{\text{total},i} - Y_i\|_2^2 = \frac{1}{5} \sum_{i=2}^{6} \|W_{\text{out}} [H_{\text{lin},i} \oplus \mathcal{NN}(H_{\text{lin},i}; \theta)] - Y_i\|_2^2$$

is used to compute the gradients with respect to both $\theta$.

After completing training, we evaluate the model using the remaining points, referred to as test points. For each prediction, we use the trained $W_{\text{out}}$ and $H_{\text{total},i}$.

$$\hat{Y}_7 = W_{\text{out}} H_{\text{total},7}, \text{ where } H_{\text{total},7} = H_{\text{lin},7} \oplus H_{\mathcal{NN},7}$$
$$\hat{Y}_8 = W_{\text{out}} H_{\text{total},8}, \text{ where } H_{\text{total},8} = H_{\text{lin},8} \oplus H_{\mathcal{NN},8}$$

and so on.

## Supplementary Tables

| **Model** | **Hyperparameter** | **Value** |
|---|---|---|
| Standard NVAR | Skipping parameter ($s$) | 1 |
| Adaptive NVAR | Skipping parameter ($s$) | 1 |
| Adaptive NVAR (Lorenz–96) | Mini-batch size | 1024 |
| ESN | Washout length (test/validation) | 500 (MG), 100 (L63) |
| HESN | Washout length (test/validation) | 500 (MG), 100 (L63) |

Supplementary Table 3: Fixed hyperparameters used in each model.

| Noise Level | Dataset | Horizon 25 | Horizon 50 | Horizon 75 | Horizon 100 |
|---|---|---|---|---|---|
| **Adaptive NVAR** | | | | | |
| Noise free | Shared (All) | $0.0057 \pm 0.0015$ | $0.0079 \pm 0.0025$ | $0.0103 \pm 0.0034$ | $0.0133 \pm 0.0047$ |
| Low noise | Dataset 1 | $0.0114 \pm 0.0004$ | $0.0138 \pm 0.0006$ | $0.0163 \pm 0.0008$ | $0.0195 \pm 0.0012$ |
| | Dataset 2 | $0.0135 \pm 0.0006$ | $0.0170 \pm 0.0010$ | $0.0206 \pm 0.0138$ | $0.0249 \pm 0.0017$ |
| | Dataset 3 | $0.0122 \pm 0.0004$ | $0.0152 \pm 0.0007$ | $0.0187 \pm 0.0011$ | $0.0223 \pm 0.0013$ |
| Moderate noise | Dataset 1 | $0.0225 \pm 0.0006$ | $0.0271 \pm 0.0007$ | $0.0326 \pm 0.0010$ | $0.0389 \pm 0.0012$ |
| | Dataset 2 | $0.0214 \pm 0.0004$ | $0.0252 \pm 0.0005$ | $0.0306 \pm 0.0006$ | $0.0387 \pm 0.0011$ |
| | Dataset 3 | $0.0220 \pm 0.0006$ | $0.0261 \pm 0.0007$ | $0.0304 \pm 0.0009$ | $0.0349 \pm 0.0014$ |
| High noise | Dataset 1 | $0.0304 \pm 0.0005$ | $0.0375 \pm 0.0005$ | $0.0455 \pm 0.0007$ | $0.0540 \pm 0.0010$ |
| | Dataset 2 | $0.0321 \pm 0.0004$ | $0.0390 \pm 0.0004$ | $0.0456 \pm 0.0006$ | $0.0530 \pm 0.0009$ |
| | Dataset 3 | $0.0329 \pm 0.0003$ | $0.0404 \pm 0.0007$ | $0.0479 \pm 0.0008$ | $0.0550 \pm 0.0008$ |
| **NG-RC** | | | | | |
| Noise free | Shared (All) | 0.0385 | 0.0529 | 0.0636 | 0.0729 |
| Low noise | Dataset 1 | 0.0240 | 0.0318 | 0.0390 | 0.0471 |
| | Dataset 2 | 0.0253 | 0.0320 | 0.0389 | 0.0476 |
| | Dataset 3 | 0.0256 | 0.0331 | 0.0397 | 0.0479 |
| Moderate noise | Dataset 1 | 0.0340 | 0.0427 | 0.0537 | 0.0669 |
| | Dataset 2 | 0.0329 | 0.0404 | 0.0504 | 0.0637 |
| | Dataset 3 | 0.0345 | 0.0434 | 0.0548 | 0.0679 |
| High noise | Dataset 1 | 0.0459 | 0.0573 | 0.0697 | 0.0833 |
| | Dataset 2 | 0.0510 | 0.0579 | 0.0668 | 0.0770 |
| | Dataset 3 | 0.0511 | 0.0641 | 0.0762 | 0.0884 |
| **ESN** | | | | | |
| Noise free | Shared (All) | $0.0321 \pm 0.0006$ | $0.0319 \pm 0.0010$ | $0.0318 \pm 0.0015$ | $0.0318 \pm 0.0018$ |
| Low noise | Dataset 1 | $0.0404 \pm 0.0223$ | $0.0432 \pm 0.0283$ | $0.0459 \pm 0.0319$ | $0.0497 \pm 0.0350$ |
| | Dataset 2 | $0.0374 \pm 0.0100$ | $0.0420 \pm 0.0204$ | $0.0472 \pm 0.0302$ | $0.0535 \pm 0.0377$ |
| | Dataset 3 | $0.0532 \pm 0.0321$ | $0.0611 \pm 0.0391$ | $0.0759 \pm 0.0541$ | $0.0861 \pm 0.0634$ |
| Moderate noise | Dataset 1 | $0.0330 \pm 0.0024$ | $0.0348 \pm 0.0041$ | $0.0371 \pm 0.0064$ | $0.0418 \pm 0.0097$ |
| | Dataset 2 | $0.0365 \pm 0.0024$ | $0.0365 \pm 0.0036$ | $0.0375 \pm 0.0052$ | $0.0411 \pm 0.0078$ |
| | Dataset 3 | $0.0587 \pm 0.0105$ | $0.0784 \pm 0.0179$ | $0.0928 \pm 0.0202$ | $0.1035 \pm 0.0188$ |
| High noise | Dataset 1 | $0.0356 \pm 0.0028$ | $0.0385 \pm 0.0042$ | $0.0435 \pm 0.0055$ | $0.0500 \pm 0.0074$ |
| | Dataset 2 | $0.0367 \pm 0.0059$ | $0.0418 \pm 0.0087$ | $0.0479 \pm 0.0114$ | $0.0567 \pm 0.0144$ |
| | Dataset 3 | $0.0540 \pm 0.0191$ | $0.0676 \pm 0.0357$ | $0.0768 \pm 0.0444$ | $0.0847 \pm 0.0484$ |
| **HESN** | | | | | |
| Noise free | Shared (All) | $0.0390 \pm 0.0093$ | $0.0470 \pm 0.0217$ | $0.0559 \pm 0.0397$ | $0.0686 \pm 0.0684$ |
| Low noise | Dataset 1 | $0.0336 \pm 0.0022$ | $0.0349 \pm 0.0035$ | $0.0362 \pm 0.0047$ | $0.0386 \pm 0.0061$ |
| | Dataset 2 | $0.0365 \pm 0.0144$ | $0.0394 \pm 0.0181$ | $0.0420 \pm 0.0213$ | $0.0451 \pm 0.0237$ |
| | Dataset 3 | $0.0338 \pm 0.0040$ | $0.0353 \pm 0.0053$ | $0.0367 \pm 0.0059$ | $0.0387 \pm 0.0066$ |
| Moderate noise | Dataset 1 | $0.0383 \pm 0.0077$ | $0.0435 \pm 0.0109$ | $0.0497 \pm 0.0150$ | $0.0567 \pm 0.0184$ |
| | Dataset 2 | $0.0340 \pm 0.0010$ | $0.0362 \pm 0.0021$ | $0.0386 \pm 0.0032$ | $0.0421 \pm 0.0045$ |
| | Dataset 3 | $0.0341 \pm 0.0071$ | $0.0371 \pm 0.0100$ | $0.0407 \pm 0.0127$ | $0.0449 \pm 0.0137$ |
| High noise | Dataset 1 | $0.0395 \pm 0.0063$ | $0.0446 \pm 0.0077$ | $0.0510 \pm 0.0102$ | $0.0581 \pm 0.0124$ |
| | Dataset 2 | $0.0364 \pm 0.0055$ | $0.0385 \pm 0.0077$ | $0.0414 \pm 0.0106$ | $0.0452 \pm 0.0126$ |
| | Dataset 3 | $0.0445 \pm 0.0159$ | $0.0501 \pm 0.0214$ | $0.0556 \pm 0.0252$ | $0.0610 \pm 0.0278$ |

Supplementary Table 4: Robustness analysis of Adaptive NVAR, NG-RC, ESN, and HESN models evaluated on three distinct Mackey-Glass datasets. Horizons 25–100 are reported across four noise regimes (Noise-free, Low, Moderate, and High). For stochastic models, results indicate mean ± standard deviation. NG-RC performance is reported as a scalar value due to its deterministic execution. (Rounded)

| Noise Level | Dataset | Horizon 25 | Horizon 50 | Horizon 75 | Horizon 100 |
|---|---|---|---|---|---|
| **Adaptive NVAR** | | | | | |
| Noise free | Shared (All) | $0.0064 \pm 0.0009$ | $0.0166 \pm 0.0039$ | $0.0303 \pm 0.0088$ | $0.0610 \pm 0.0306$ |
| Low noise | Dataset 1 | $0.0915 \pm 0.0117$ | $0.1132 \pm 0.0106$ | $0.1297 \pm 0.0120$ | $0.1613 \pm 0.0195$ |
| | Dataset 2 | $0.0787 \pm 0.0034$ | $0.1009 \pm 0.0037$ | $0.1292 \pm 0.0082$ | $0.1703 \pm 0.0126$ |
| | Dataset 3 | $0.0889 \pm 0.0028$ | $0.1148 \pm 0.0053$ | $0.1457 \pm 0.0070$ | $0.2410 \pm 0.0148$ |
| Moderate noise | Dataset 1 | $0.1541 \pm 0.0033$ | $0.1906 \pm 0.0031$ | $0.2874 \pm 0.0124$ | $0.3730 \pm 0.0118$ |
| | Dataset 2 | $0.1435 \pm 0.0038$ | $0.1799 \pm 0.0094$ | $0.2457 \pm 0.0175$ | $0.3775 \pm 0.0244$ |
| | Dataset 3 | $0.2331 \pm 0.0117$ | $0.2777 \pm 0.0139$ | $0.3651 \pm 0.0327$ | $0.4624 \pm 0.0428$ |
| High noise | Dataset 1 | $0.3305 \pm 0.0156$ | $0.3636 \pm 0.0194$ | $0.4408 \pm 0.0241$ | $0.5160 \pm 0.0297$ |
| | Dataset 2 | $0.2330 \pm 0.0128$ | $0.3196 \pm 0.0178$ | $0.3992 \pm 0.0392$ | $0.4768 \pm 0.0413$ |
| | Dataset 3 | $0.3216 \pm 0.0062$ | $0.3747 \pm 0.0210$ | $0.4883 \pm 0.0275$ | $0.6110 \pm 0.0367$ |
| **NG-RC** | | | | | |
| Noise free | Shared (All) | 0.0023 | 0.0054 | 0.0081 | 0.0102 |
| Low noise | Dataset 1 | 0.1140 | 0.2474 | 0.4119 | 0.5295 |
| | Dataset 2 | 0.1589 | 0.2501 | 0.3794 | 0.5199 |
| | Dataset 3 | 0.1017 | 0.1795 | 0.2646 | 0.3219 |
| Moderate noise | Dataset 1 | 0.2267 | 0.3103 | 0.4653 | 0.6536 |
| | Dataset 2 | 0.1631 | 0.3108 | 0.4590 | 0.5986 |
| | Dataset 3 | 0.2428 | 0.3691 | 0.5460 | 0.6981 |
| High noise | Dataset 1 | 0.2546 | 0.4216 | 0.6366 | 0.8062 |
| | Dataset 2 | 0.3041 | 0.4861 | 0.6913 | 0.8619 |
| | Dataset 3 | 0.2956 | 0.5347 | 0.7518 | 0.9183 |
| **ESN** | | | | | |
| Noise free | Shared (All) | $0.1514 \pm 0.0040$ | $0.1944 \pm 0.0379$ | $0.2852 \pm 0.0934$ | $0.4207 \pm 0.1547$ |
| Low noise | Dataset 1 | $0.1525 \pm 0.0072$ | $0.1724 \pm 0.0170$ | $0.2077 \pm 0.0302$ | $0.2507 \pm 0.0432$ |
| | Dataset 2 | $0.1658 \pm 0.0118$ | $0.2106 \pm 0.0207$ | $0.2572 \pm 0.0297$ | $0.3186 \pm 0.0422$ |
| | Dataset 3 | $0.1773 \pm 0.0134$ | $0.2125 \pm 0.0204$ | $0.2525 \pm 0.0353$ | $0.3500 \pm 0.0390$ |
| Moderate noise | Dataset 1 | $0.1987 \pm 0.0147$ | $0.2893 \pm 0.0381$ | $0.3931 \pm 0.0610$ | $0.4955 \pm 0.0789$ |
| | Dataset 2 | $0.1853 \pm 0.0118$ | $0.2403 \pm 0.0324$ | $0.3766 \pm 0.0603$ | $0.5044 \pm 0.0925$ |
| | Dataset 3 | $0.2127 \pm 0.0212$ | $0.3151 \pm 0.0347$ | $0.4200 \pm 0.0630$ | $0.5281 \pm 0.0854$ |
| High noise | Dataset 1 | $0.2808 \pm 0.0251$ | $0.4061 \pm 0.0504$ | $0.5894 \pm 0.0748$ | $0.7412 \pm 0.0831$ |
| | Dataset 2 | $0.2123 \pm 0.0255$ | $0.3419 \pm 0.0507$ | $0.4400 \pm 0.0666$ | $0.5280 \pm 0.0691$ |
| | Dataset 3 | $0.2466 \pm 0.0172$ | $0.3296 \pm 0.0286$ | $0.4625 \pm 0.0407$ | $0.6065 \pm 0.0565$ |
| **HESN** | | | | | |
| Noise free | Shared (All) | $0.1965 \pm 0.0106$ | $0.3008 \pm 0.0406$ | $0.5066 \pm 0.0570$ | $0.6384 \pm 0.0795$ |
| Low noise | Dataset 1 | $0.1721 \pm 0.0043$ | $0.2456 \pm 0.0175$ | $0.3904 \pm 0.0340$ | $0.4967 \pm 0.0563$ |
| | Dataset 2 | $0.1772 \pm 0.0140$ | $0.2236 \pm 0.0186$ | $0.2846 \pm 0.0354$ | $0.3541 \pm 0.0534$ |
| | Dataset 3 | $0.1784 \pm 0.0149$ | $0.2123 \pm 0.0258$ | $0.2668 \pm 0.0358$ | $0.3549 \pm 0.0490$ |
| Moderate noise | Dataset 1 | $0.1809 \pm 0.0108$ | $0.2711 \pm 0.0363$ | $0.3772 \pm 0.0724$ | $0.5122 \pm 0.0952$ |
| | Dataset 2 | $0.2061 \pm 0.0173$ | $0.3647 \pm 0.0594$ | $0.5548 \pm 0.0738$ | $0.7211 \pm 0.0859$ |
| | Dataset 3 | $0.2334 \pm 0.0197$ | $0.3615 \pm 0.0588$ | $0.5230 \pm 0.1041$ | $0.6704 \pm 0.1206$ |
| High noise | Dataset 1 | $0.2828 \pm 0.0360$ | $0.4290 \pm 0.0740$ | $0.6075 \pm 0.1096$ | $0.7367 \pm 1.1151$ |
| | Dataset 2 | $0.2485 \pm 0.0282$ | $0.4270 \pm 0.0784$ | $0.5763 \pm 0.1044$ | $0.6854 \pm 0.1208$ |
| | Dataset 3 | $0.3422 \pm 0.0185$ | $0.5651 \pm 0.0477$ | $0.7778 \pm 0.0702$ | $0.8882 \pm 0.0726$ |

Supplementary Table 5: Robustness analysis of Adaptive NVAR, NG-RC, ESN, and HESN models evaluated on three distinct Lorenz 63 datasets. Horizons 25–100 are reported across four noise regimes (Noise-free, Low, Moderate, and High). For stochastic models, results indicate mean ± standard deviation. NG-RC performance is reported as a scalar value due to its deterministic execution. (Rounded)

| Dataset | Noise Level | 25 Steps | 50 Steps | 75 Steps | 100 Steps |
|---|---|---|---|---|---|
| Dataset 1 | Noise free | $0.0306 \pm 0.0040$ | $0.0612 \pm 0.0091$ | $0.1029 \pm 0.0158$ | $0.1597 \pm 0.0236$ |
| | Low noise | $0.0859 \pm 0.0011$ | $0.1475 \pm 0.0023$ | $0.2286 \pm 0.0036$ | $0.3254 \pm 0.0048$ |
| | Moderate noise | $0.1684 \pm 0.0056$ | $0.2719 \pm 0.0120$ | $0.3873 \pm 0.0162$ | $0.5009 \pm 0.0173$ |
| | High noise | $0.2367 \pm 0.0015$ | $0.3664 \pm 0.0034$ | $0.5004 \pm 0.0047$ | $0.6172 \pm 0.0053$ |
| Dataset 2 | Noise free | $0.0306 \pm 0.0040$ | $0.0612 \pm 0.0091$ | $0.1029 \pm 0.0158$ | $0.1597 \pm 0.0236$ |
| | Low noise | $0.0854 \pm 0.0009$ | $0.1464 \pm 0.0019$ | $0.2270 \pm 0.0030$ | $0.3232 \pm 0.0040$ |
| | Moderate noise | $0.1682 \pm 0.0050$ | $0.2712 \pm 0.0109$ | $0.3862 \pm 0.0149$ | $0.4998 \pm 0.0161$ |
| | High noise | $0.2357 \pm 0.0010$ | $0.3642 \pm 0.0023$ | $0.4976 \pm 0.0036$ | $0.6145 \pm 0.0045$ |
| Dataset 3 | Noise free | $0.0306 \pm 0.0040$ | $0.0612 \pm 0.0091$ | $0.1029 \pm 0.0158$ | $0.1597 \pm 0.0236$ |
| | Low noise | $0.0858 \pm 0.0013$ | $0.1474 \pm 0.0027$ | $0.2287 \pm 0.0041$ | $0.3254 \pm 0.0053$ |
| | Moderate noise | $0.1693 \pm 0.0063$ | $0.2733 \pm 0.0136$ | $0.3892 \pm 0.0183$ | $0.5029 \pm 0.0196$ |
| | High noise | $0.2366 \pm 0.0014$ | $0.3664 \pm 0.0031$ | $0.5005 \pm 0.0046$ | $0.6180 \pm 0.0058$ |

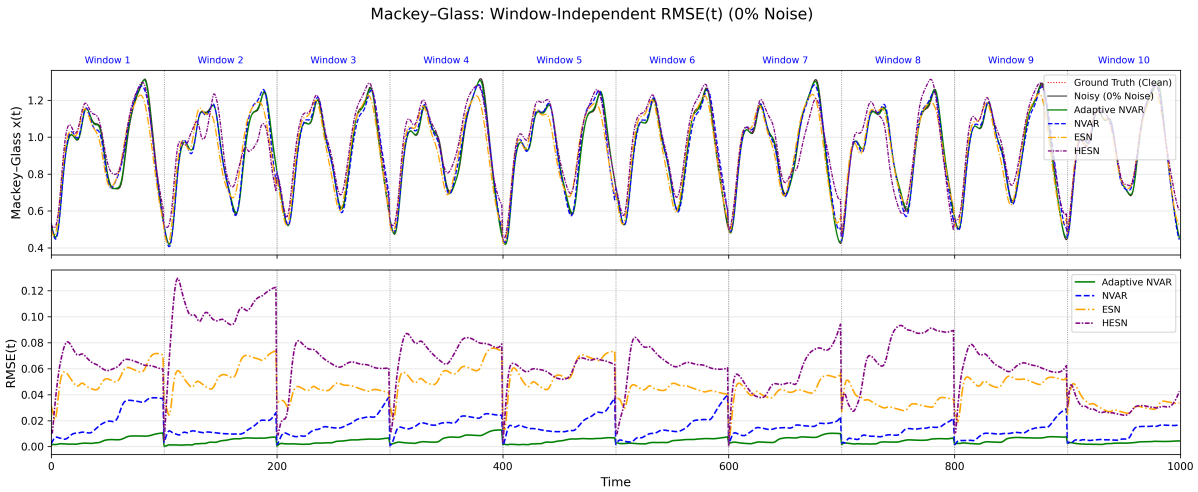
Supplementary Table 6: Forecasting RMSE (mean ± standard deviation) of the Adaptive NVAR model under different noise levels and datasets on Lorenz 96 ($N = 100$).

| Model | System | Feature Formula | Parameter Values | Feature Size ($m$) |
|---|---|---|---|---|
| Standard NVAR | Mackey–Glass | $dk(dk+1)/2$ | $d = 1$, $k = 10$ | 55 |
| Standard NVAR | Lorenz–63 | $dk(dk+1)/2$ | $d = 3$, $k = 10$ | 4,515 |
| Standard NVAR | Lorenz–96 | $dk(dk+1)/2$ | $d = 100$, $k = 20$ | 2,001,000 |
| Adaptive NVAR | Mackey–Glass | $dk(dk+1)/2$ | $d = 1$, $k = 10$ | 55 |
| Adaptive NVAR | Lorenz–63 | $dk(dk+1)/2$ | $d = 3$, $k = 10$ | 4,515 |
| Adaptive NVAR | Lorenz–96 | $c\,dk$ | $n = 10$, $d = 100$, $k = 20$ | 20,000 |

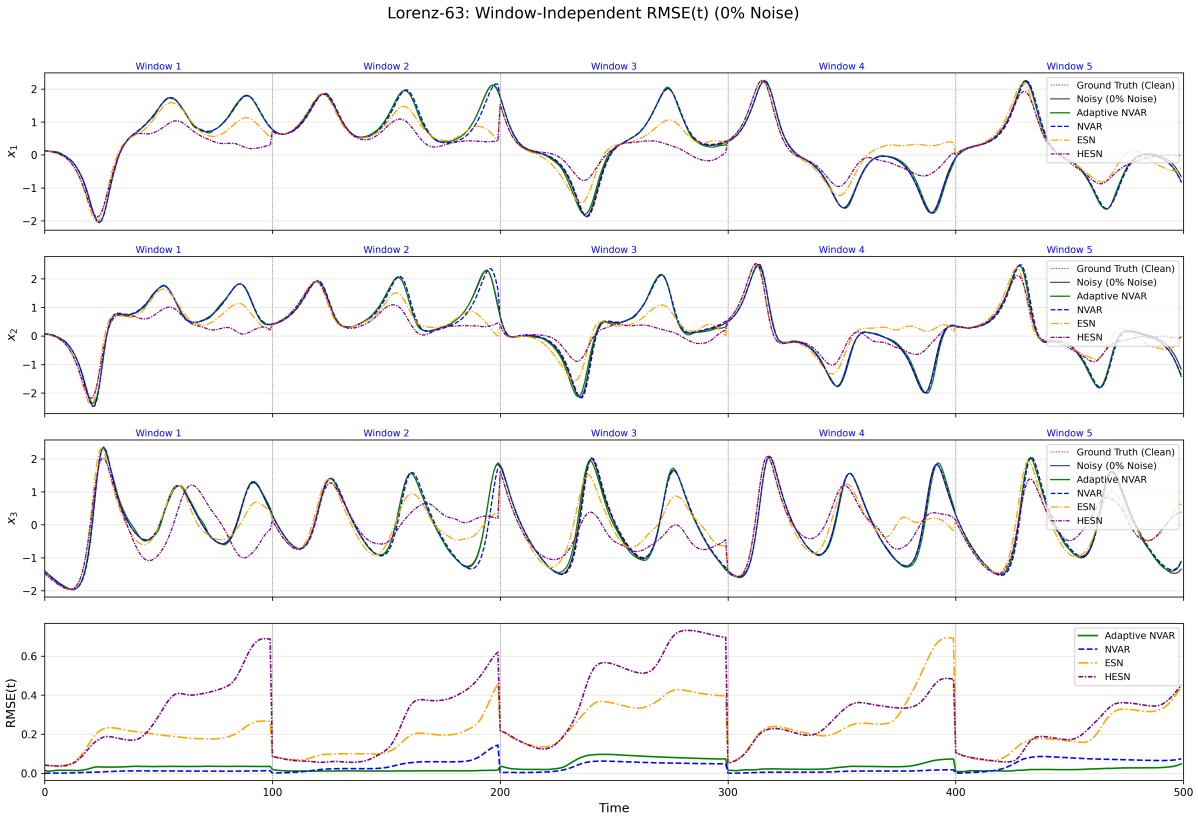
Supplementary Table 7: Examples: Nonlinear feature dimensionality for the NVAR and Adaptive NVAR models . Here, $d$ is the system dimension, $k$ the number of delay taps, and $n$ the number of hidden neurons per delayed variable in the Adaptive NVAR. For high-dimensional systems such as Lorenz–96, the Adaptive NVAR replaces the polynomial feature expansion with a shallow MLP of output size $m = n\,dk$.

# Supplementary Figures

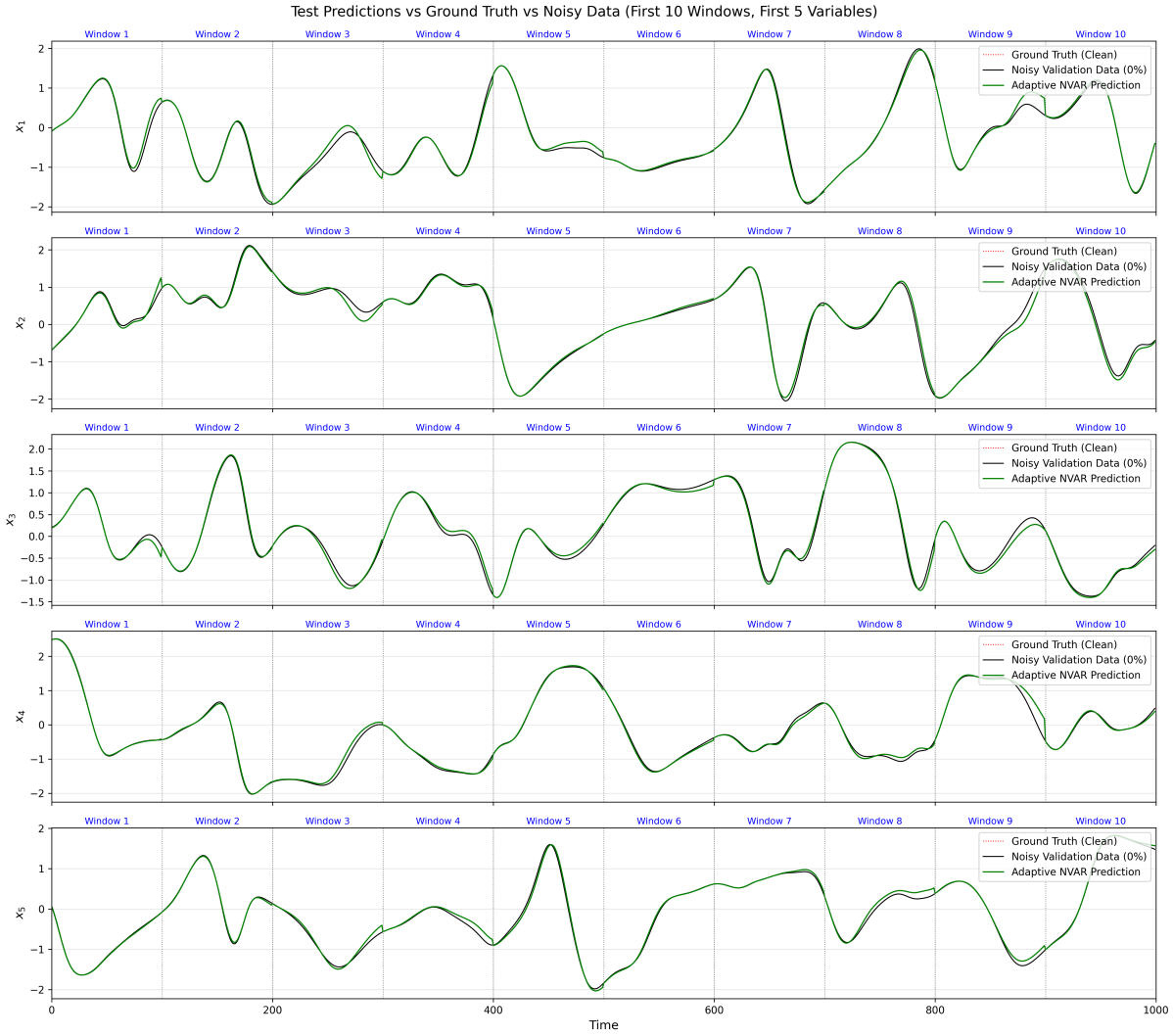
## Noise-Free (0%)




Supplementary Figure 1: **Forecasting performance of the Mackey–Glass system (0% noise).** Top panel shows the trajectories of the true signal (red, dotted), noisy input (black), and model predictions (colored lines). Bottom panel shows the window-wise RMSE(t) for non-overlapping windows of length 100.

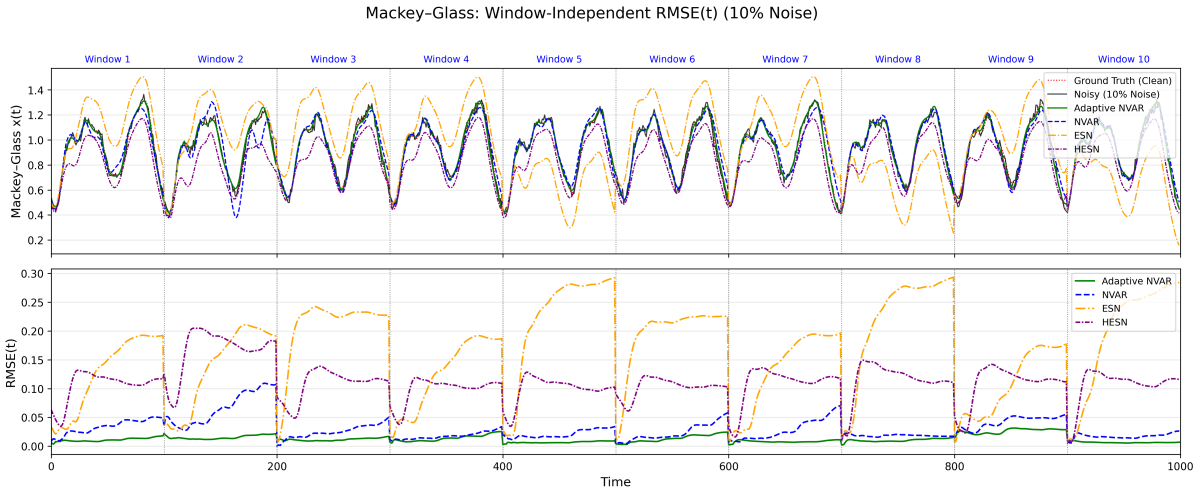



Supplementary Figure 2: **Forecasting performance of the Lorenz–63 system (0% noise).** Each subplot shows the true signal (red, dotted), noisy input (black), and predictions (colored lines) for each state variable. Vertical dotted lines indicate forecast windows of 100 time steps used to compute window-wise RMSE.

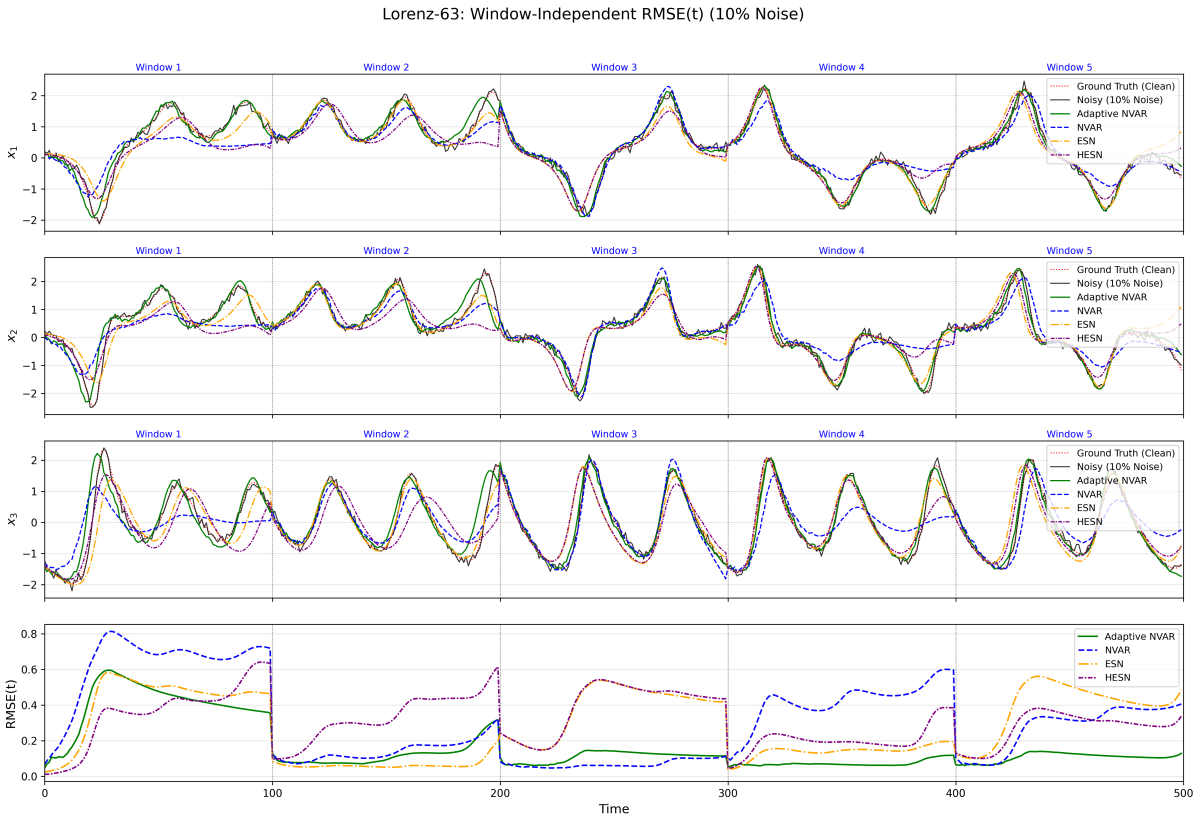



Supplementary Figure 3: **Forecasting performance of the high-dimensional Lorenz–96 system (0% noise).** True signal (red, dotted), noisy input (black), and Adaptive NVAR predictions (green) for the first 5 variables and first 10 forecast windows (each of 100 time steps).

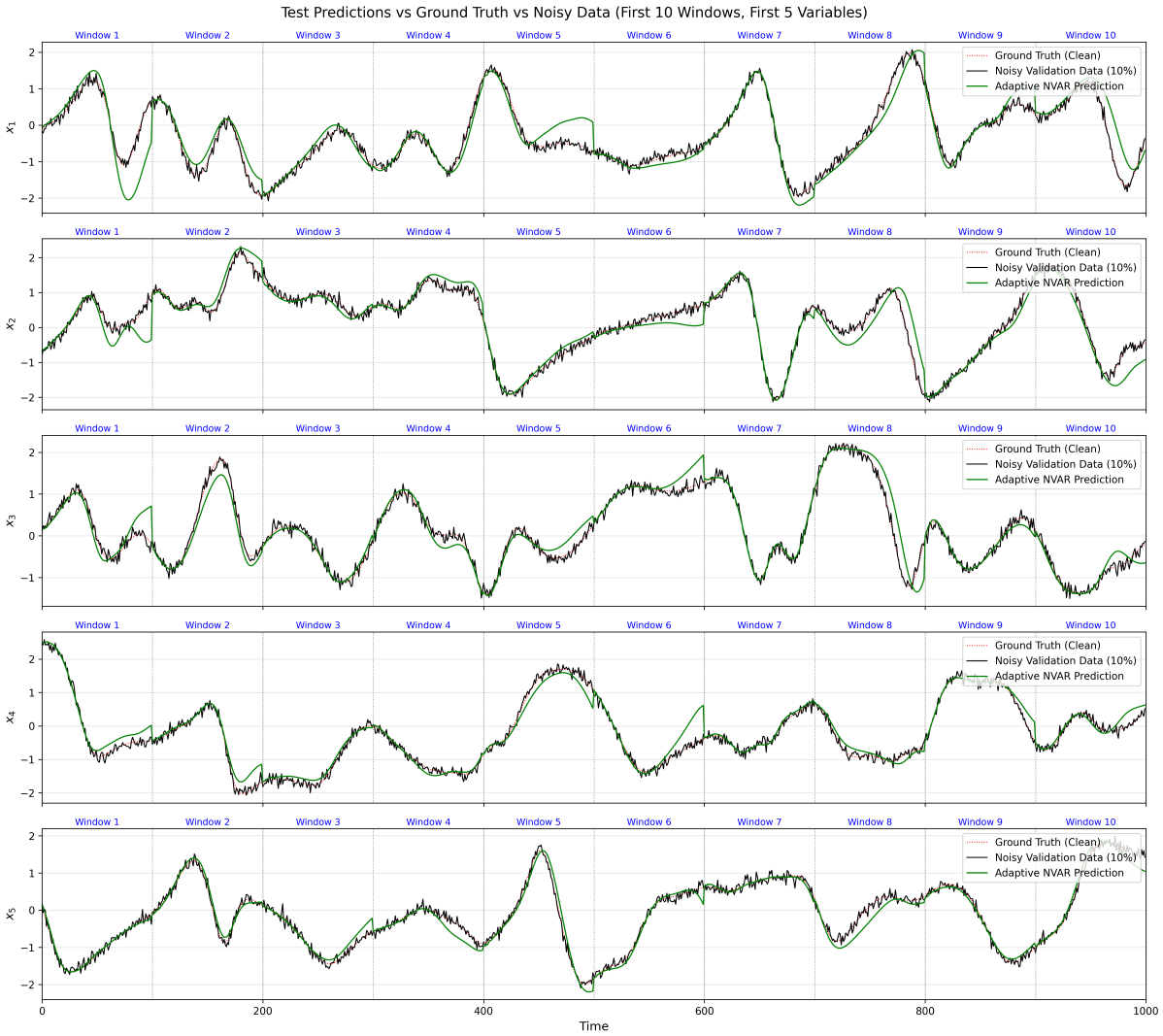
## Low Noise (10%)




Supplementary Figure 4: **Forecasting performance of the Mackey–Glass system (10% noise).**

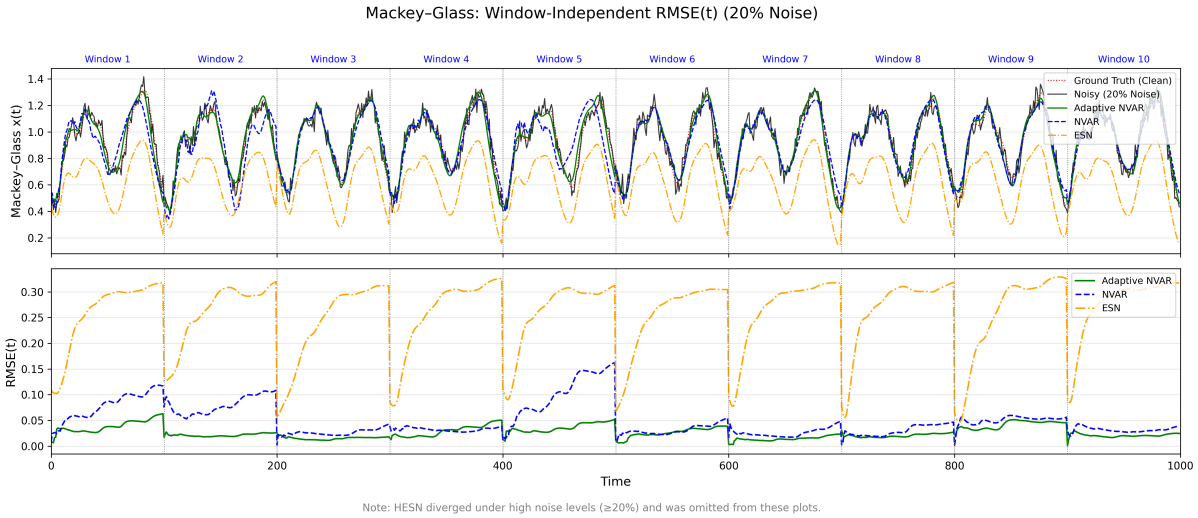



Supplementary Figure 5: **Forecasting performance of the Lorenz–63 system (10% noise).**

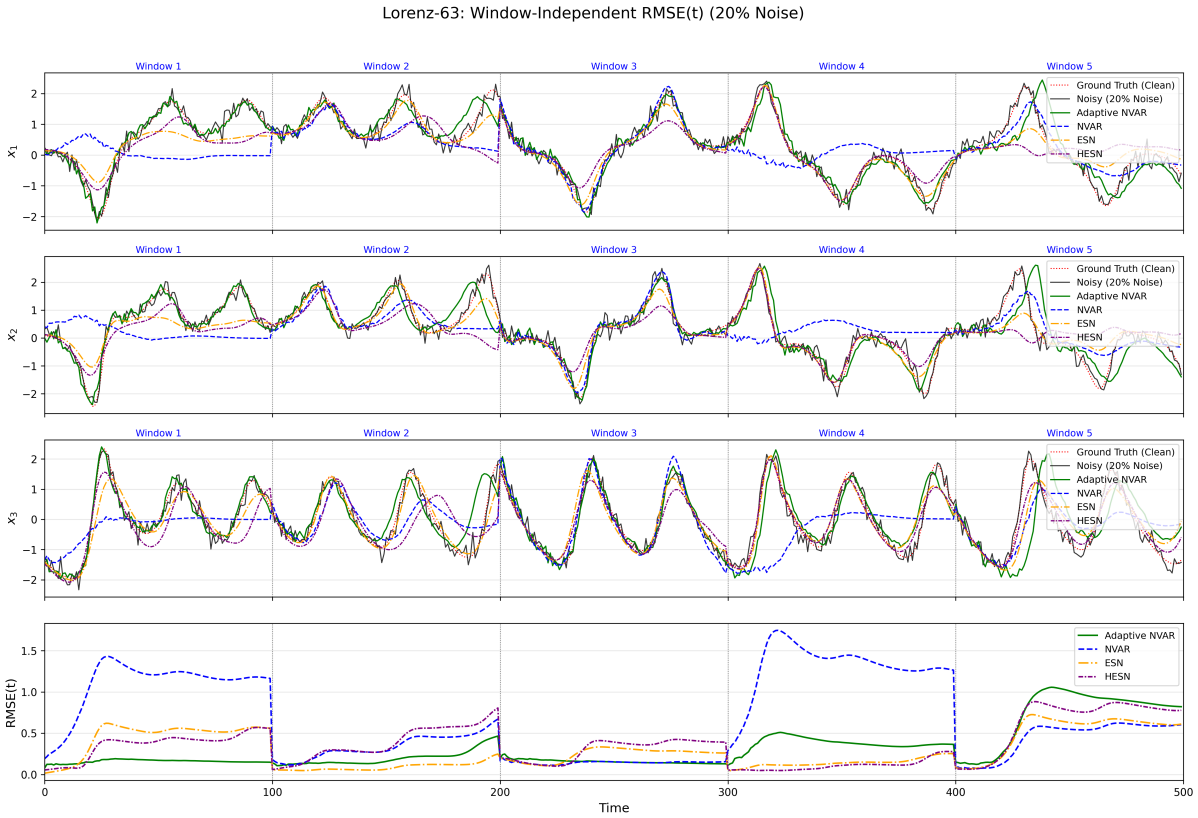



Supplementary Figure 6: **Forecasting performance of the high-dimensional Lorenz–96 system (10% noise).**

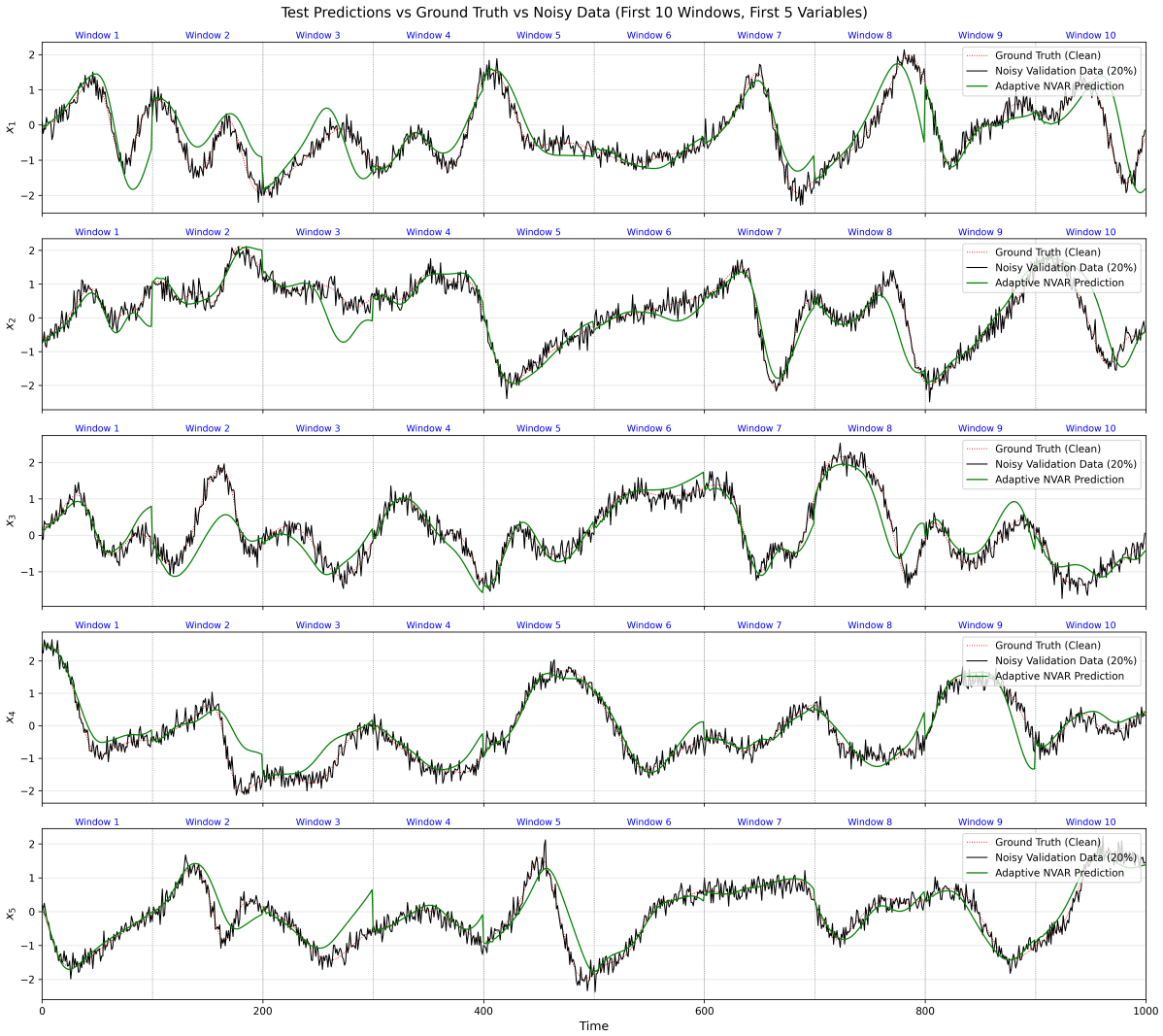
## Moderate Noise (20%)




Supplementary Figure 7: **Forecasting performance of the Mackey–Glass system (20% noise).**




Supplementary Figure 8: **Forecasting performance of the Lorenz–63 system (20% noise).**



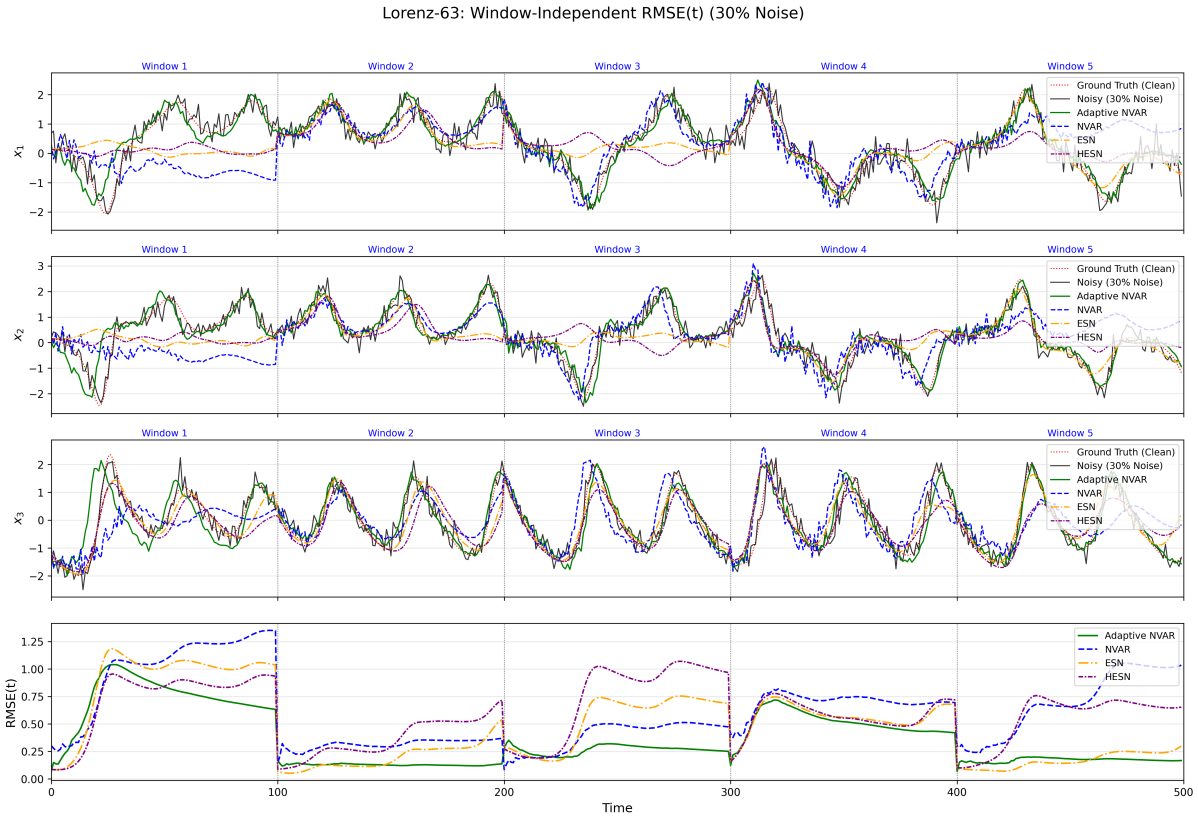

Supplementary Figure 9: **Forecasting performance of the high-dimensional Lorenz–96 system (20% noise).**

## High Noise (30%)

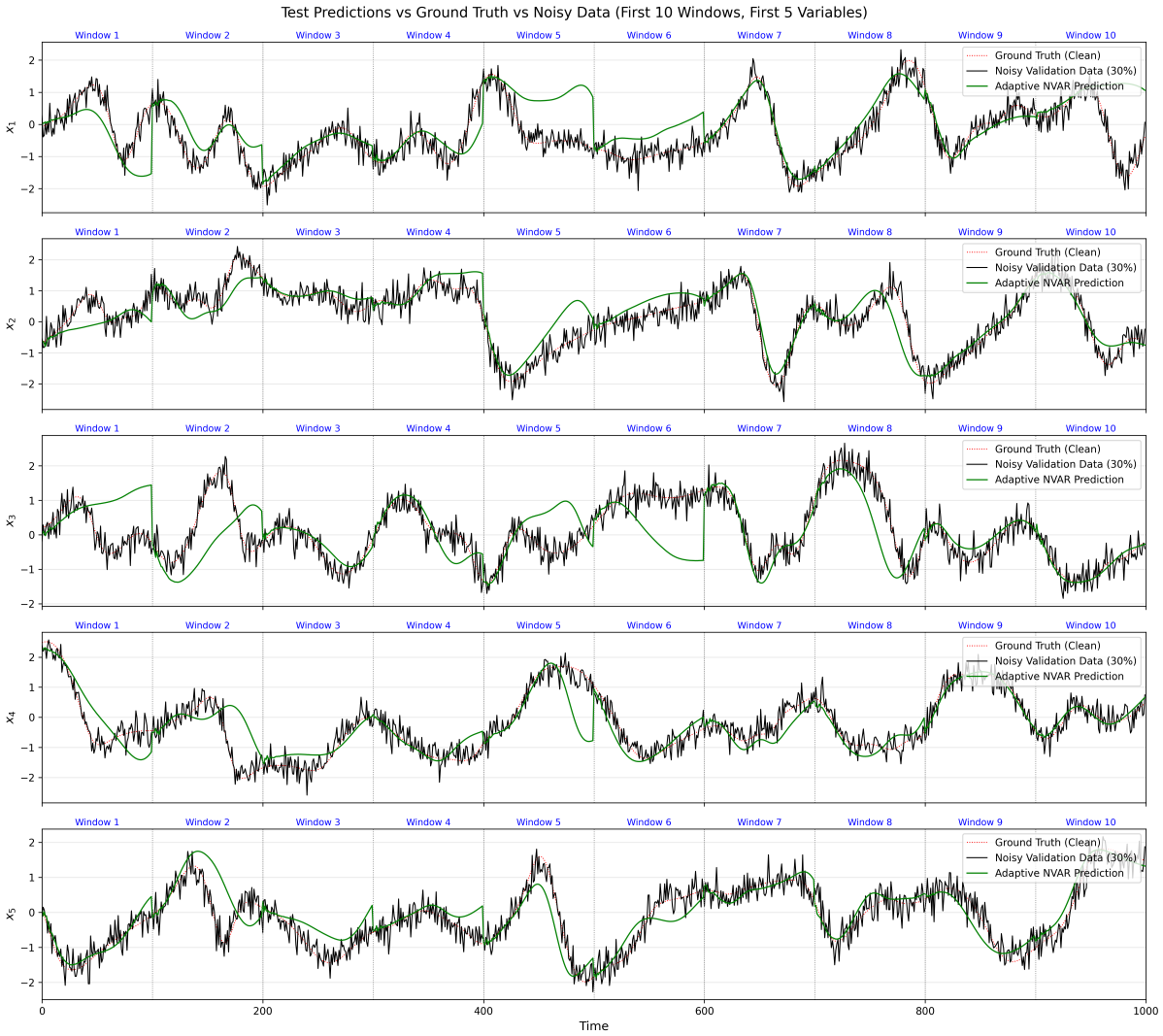



Note: HESN diverged under high noise levels (≥20%) and was omitted from these plots.

Supplementary Figure 10: **Forecasting performance of the Mackey–Glass system (30% noise).**




Supplementary Figure 11: **Forecasting performance of the Lorenz–63 system (30% noise).**




Supplementary Figure 12: **Forecasting performance of the high-dimensional Lorenz–96 system (30% noise).**

## Note on hyperparameter search strategies

The hyperparameter search intervals represent the complete ranges explored across all benchmark systems for all models being evaluated. To ensure computational feasibility and to adapt to the distinct dynamics of each system, system-specific subsets were applied:

- **Mackey–Glass:** Lightweight 1D dynamics allowed a broad exploration of hidden dimensions and learning rates ($k \in \{2, 10, 30, 50\}$, hidden units $\in \{10, 20, 50, 100, 200, 500, 1000\}$, Adam learning rate $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$, L-BFGS learning rate $\in \{1.0, 0.5, 0.1\}$).
- **Lorenz–63:** To capture moderate chaotic complexity, higher hidden dimensions were used for stable optimization ($k \in \{2, 10, 30, 50\}$, hidden units $\in \{100, 500, 1000\}$) and smaller learning rates ($\{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$).
- **Lorenz–96:** For high-dimensional settings ($d = 100$), computational scalability was the limiting factor. Thus, a reduced grid was employed ($k \in \{2, 10, 30\}$, hidden units $\in \{200, 500, 1000\}$, Adam learning rate $\in \{10^{-3}, 10^{-4}, 10^{-5}\}$).